How Much Data Do Transformers Need? A Scaling Analysis of DistilBERT for Fake News Detection

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Abstract. Online fake news is not of recent origin, but social media has provided it with an unexampled magnitude and speed. With an increasing number of individuals using these sites as sources of news, there has been an increased need and difficulty in distinguishing between authentic journalism and fake news. The three generations of NLP models considered in this paper are classical machine learning methods (Logistic Regression, SVM, and Random Forest using TF-IDF features), a sequential deep learning model (LSTM), and a transformer (DistilBERT). Each of them is trained and tested on WELFake, 71,537 annotated news articles, in an 80/20 split. This paper examines the behavior of DistilBERT as the size of training data is scaled up, will not rely on the assumption that large datasets are available. The model is trained with three data sets: 1000, 3000 and 5000 samples, and the setup is restricted to CPU-based execution. The concept is quite straightforward: the vast majority of published findings are based on the use of powerful hardware and complete datasets which are not always accessible in the actual academic environment. On the contrary, this paper is concerned with what is feasible within such limitations. As a benchmark, the baseline models are already scoring high with SVM at 96.46 and LSTM at a bit higher 96.55. A definite trend is observed when DistilBERT is trained on smaller subsets. With 1000 samples, it achieves 94.41%, which then rises noticeably to 96.00% at 3000 samples, and only slightly to 96.07% at 5000 samples. What is interesting is that the 3000-sample size is equal to only 5.2% of the entire training data (57,229 samples on which the SVM is trained), but the performance difference is very small (0.46%). Practically, the majority of the gains occur at the beginning, between 1000 and 3000 samples, beyond which the marginal gains are slight. This implies that despite the scarcity of data, even transformer architectures can learn the essential patterns needed to perform the task, and therefore they can be more readily available than it would be thought in low-resource settings. The other finding of this analysis is the importance of preprocessing that is commonly considered as an opportunity cost in most articles. Even the very simple options like the size of the vocabulary, n-gram structure and the removal of stopwords can have a significant impact on classical model performance, even more than the algorithm itself. These steps are not always reported clearly hence it becomes hard to reproduce results. The fact that they had such an influence is important to emphasize the need to plan preprocessing carefully, particularly when using limited resources. These findings will provide the first academic resource constrained deployment guidelines for transformers on WELFake.

Keywords: Fake News Detection , NLP , Machine Learning , Deep Learning , Transformers , WELFake , LSTM , DistilBERT , Data Scaling , CPU Constraints , Resource Efficient NLP

1. INTRODUCTION

Fake news is harmful. It has influenced how people reacted to health crises, influenced elections, shook financial markets, and failed institutions. It has caused many social and political problems which urge us to reliable detection systems[13]. The COVID-19 epidemic brought it to life: false information about vaccines and treatments was spread more efficiently than the official one, which had significant behavioral consequences. To a great extent, the reason as to why this occurs is structural [6,9]. Engagement in social media is rewarded by the algorithms, and sensational or emotionally-charged content is likely to be more successful than reliable but restrained reporting [6].

Human fact-checkers are useful but cannot be scaled to the amount of content being created on a daily basis. That has driven scientists in the direction of automated detection, and NLP has become the focus of most serious attempts. The space has gone through three wide stages including initial rule-based and statistical approaches, neural sequence models, and the transformer age that we are in [3, 10].

Classical models such as Logistic Regression and SVM are quick, interpretable and can be deployed with little infrastructure. FastText is used for text classification as it is both simple and efficient[17]. Their disadvantage is that bag-of-words representations completely strip away word order and word context. To a certain degree, LSTMs address this issue because they operate on a sequence of text, which enables them to identify patterns that have a multivocal scope. Such transformers as BERT are more advanced. Though XLNet can outperform BERT on various language understanding tasks[18]. ALBERT maintains strong performance by reducing various parameters[19]. Further ELECTRA's efficiency is achieved by a prediction-based approach[20]. Attention mechanisms enable them to balance relationships among all word pairs in a sentence at any given time and come up with representations that are contextually sensitive [3, 11].

The paper analyzes these three generations of models in a uniform experimental context, and also generalizes the analysis in order to interpret the way transformer based models perform under realistic limitations. In practice, high-end computational devices like GPUs are not easily accessible in a real academic setting and this has a direct impact on the feasibility of training and testing such models. This study gives more emphasis on the change in behavior of the model under limited training conditions as opposed to performance comparison. Specifically, it presents a point of view of data scaling to examine the way performance changes with the variation of the quantity of training data under such conditions.

The majority of the available research on transformer models uses full-scale datasets and trains them on GPUs, thus limiting their applicability to low-resource contexts. Relatively little work has been done to investigate the performance of these models where there are both data and computational resources constraints. In order to fill this gap, this paper performs a data scaling training of DistilBERT by using three increasing dataset sizes: 1000, 3000 and 5000 samples. All the experiments are conducted with the same settings in CPU-only execution. The aim is to determine the dependence of performance on the scale of data and to determine the minimum scale of data that can be used to obtain competitive performance. These results indicate that the transformer models are capable of working even with comparatively small data subsets, which is why they are better to fit in the limited academic and experimental contexts. A critical gap is present in existing literature that majority of transformers experiments are done with GPU hardware, which is usually not available to most academic scholars. No such systematic study has yet been done as to how DistilBERT performance scales with different dataset sizes under CPU constraints and no GPU which this paper will explore this unchartered area.

The paper compares all three generations on the same data with same division, and it tries to give reasons to its results, not merely to report them. The contributions, specific to them, are as follows:

1. A comparison of five models - Logistic Regression, SVM, Random Forest, LSTM, and DistilBERT - on the same dataset and identical conditions.
2. A data scaling analysis of DistilBERT under constrained setting, where the transformer model is trained on 1,000, 3,000, and 5,000 samples using only CPU, a study between relationship of training data size and performance.

3. A major finding that of DistilBERT trained on 3,000 samples (approximately 5.2% of the full dataset) achieves performance very close to other models trained on the entire dataset, showing the data efficiency of DistilBERT.
4. Observations on the practical behavior of preprocessing choices and their impact on the performance of classical models, which in many cases are often not declared in existing literature.

2. RELATED WORK

However, the bulk of the work in fake news detection has focused on automated methods of classification, as a result of increased interest in social media and the impact of misinformation; many recent studies have utilized computational approaches that draw upon established spam filtering techniques and sentiment analysis such as Naïve Bayes, Logistic Regression, Support Vector Machines (SVM), and Decision Trees by utilizing the Bag-of-Word or TF-IDF (Term Frequency-Inverse Document Frequency) representation of text to classify. Shu et al. [6] provide an excellent overview of the early research employing such computational methods for classifying fake news. They conclude that while these methods provided useful baseline measures for performance in detecting fake news, they were hindered by the fact that they did not encode the order of words or any semantic relationship among the words; therefore, there was a significant ceiling on the performance of these methods.

With the introduction of deep learning, CNNs and RNNs have become prominent classification algorithms as CNNs are more suitable for sentence-based classification task[15]. In particular, architectures based on Long Short-Term Memory (LSTM) networks are an attractive choice for text classification because they partially overcome the vanishing gradient problems, making it difficult to train LSTMs with standard RNNs on long input sequences [7]. In addition, several studies that compared the performance of LSTM networks with the performance of traditional classification algorithms found a significant increase in classification accuracy on fake news classification datasets. Ruchansky et al. [8] proposed a hybrid classification model that combined traditional true/false classification of article content and user engagement signals, as they felt that the propagation of fake news is a social phenomenon and linguistic phenomenon. Although progress has been made by researchers working with LSTMs, LSTMs have difficulty modeling long-range dependencies, as the network may lose important contextual information from the start of a long article before the end of the article.

Vaswani et al. [11] proposed a transformer architecture incorporating self-attention so that any position in a sequence can be related to another irrespective of how far apart the two positions are within the context of the sequence. BERT was pre-trained on large text corpora but fine-tuned for downstream tasks. By demonstrating how transfer learning can greatly diminish the amount of labeled data needed to achieve highly accurate classification, BERT has significantly changed the way NLP classification is done. Alghamdi et al. [1] used transformer-based models for the detection of misinformation related to COVID-19 and found that using this type of model achieved significant improvement when compared to traditional approaches; however, they also noted that a domain shift from general news to pandemic-related content poses some challenges. Lighter versions of transformers have continued this work. RoBERTa [4] further improved on BERT by carefully selecting data used for pretraining, creating larger batch sizes than those used in BERT, and removing next-sentence prediction as an objective during pretraining. DistilBERT [5] also improved on BERT but by distilling BERT into a model that has about half of the number of parameters of BERT but retains approximately 97% of BERT's performance on the GLUE benchmark. Hence, when working with constraints related to available resources, DistilBERT is often preferred, which is why we chose to utilize DistilBERT as part of our experiments with transformers.

The issue of dataset quality is an emerging trend seen across studies done on this data group. For example, Patwa et al. [2] published COVID-19 misinformation data and discovered that models developed from general news datasets typically do not generalize for pandemic specific misinformation; this highlights that benchmarks from one dataset are not necessarily generalizable to the real world. To address these issues, WELFake, the dataset used for this study, was created by combining four different datasets to remove source bias and create a more diverse representation of topic [12].

Another area where there is significantly less literature is with respect to how computational limitations impact published results. The majority of published benchmarks are executed on large subsets of training data using GPU

resources. This study attempts to address the gap in this literature by providing transformer results on CPU with an accordingly smaller training sample size; along with corresponding results using full datasets in both traditional and deep learning frameworks, creating better understanding of the trade-offs involved.

3. METHODOLOGY

Our experimental setup follows a standard NLP pipeline: data loading and cleaning, feature extraction, model training, and evaluation. We kept conditions as consistent as possible across all models — same dataset, same train-test split, and same random seed (42) — so that performance differences reflect the models themselves rather than experimental variation.

3.1 Dataset

We used the WELFake dataset [12], a publicly available benchmark for fake news classification constructed by merging existing news sources: Kaggle, McIntire, Reuters, and BuzzFeed Political. Existing datasets are generally limited in features and scale[14] . But WELFake raw dataset contained 72,134 articles. After removing null entries and duplicate rows, our working dataset contained 71,537 news articles labeled as either real (0) or fake (1). The class distribution is approximately balanced — in the 14,308-sample test set, 7,081 articles are labeled real, and 7,227 are labeled fake, giving a near 1:1 ratio that makes accuracy a reliable metric alongside F1-score [2].

3.2 Data Preprocessing

Article titles and body text were concatenated into a single content field, since both carry signals relevant to fake news classification. We then applied the following cleaning steps to all text: conversion to lowercase, removal of URLs, removal of non-alphabetic characters, and collapsing of extra whitespace.

For classical machine learning models, English stopwords were additionally removed to reduce noise in the TF-IDF representation. Transformer tokenization was handled by the DistilBERT tokenizer, which operates on lightly preprocessed text and manages subword segmentation internally [5].

3.3 Feature Extraction

For classical machine learning models, text was converted to numerical vectors using TF-IDF with the following configuration: vocabulary size of 10,000 features, unigram and bigram coverage (ngram_range=(1,2)), and a maximum document frequency of 0.7 to filter terms appearing in more than 70% of documents. The TF-IDF weight for term t in document d is defined as:

$$TF - IDF(t, d) = TF(t, d) * \log\left(\frac{N}{DF(t)}\right) \quad (1)$$

Where TF (t, d) is the frequency of term t in document d, DF(t) is the number of documents containing t, and N is the total number of documents. This weighting elevates terms that are informative for a specific document while downweighing very common terms. In most NLP tasks where text representation is used, word embedding is quite common[16]. The LSTM model used a learned embedding layer; DistilBERT used its pre-trained subword tokenizer with max_length=64 [3].

3.4 Machine Learning Models

Three classical models were trained:

- Logistic Regression with max_iter=1000. It predicts class probability using the sigmoid function:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad \textbf{(2)}$$

where w is the learned weight vector, x is the TF-IDF feature vector, and b is the bias term.

- Support Vector Machine (SVM) using LinearSVC with default settings. SVM finds a maximum-margin hyperplane separating the two classes in the TF-IDF feature space [6].

- Random Forest with n_estimators=50, random_state=42, and n_jobs=-1. It aggregates predictions from 50 decision trees; each trained on a random subset of features [6].

3.5 Deep Learning Model (LSTM)

The LSTM architecture consisted of the following layers: an Embedding layer (vocab_size=10000, embedding_dim=128), a single LSTM layer with 64 hidden units, a Dropout layer with rate 0.5 for regularization, and a Dense output layer with sigmoid activation for binary classification. The model was compiled with binary cross-entropy loss and the Adam optimizer, then trained for 5 epochs with batch_size=32 and a 10% validation split drawn from the training data. Input sequences were padded to a fixed length of 500 tokens.
The LSTM updates its hidden state at each time step as:

$$h_t = LSTM(h_{t-1}, x_t) \quad \textbf{(3)}$$

Where h_t is the current hidden state, h_{t-1} is the previous hidden state, and x_t is the embedded word input at step t. The internal gating mechanisms of the LSTM cell allow the model to selectively retain or discard information across the sequence [7].

3.6 Transformer Model (DistilBERT)

We optimized the distilbert-base-uncased [5] on binary sequence classification with the Hugging Face Transformers library. The 3,000-sample subsample of the training data was trained with the following settings: max_length=64 tokens, per_device_train_batch_size=2, and num_train_epochs=2 because of the CPU-only constraints in Google Colab. We have clearly stated such circumstances as these do have a direct impact on the results and can be relevant to any reader working under the same resource constraints.

To further scale this setup, we also train DistilBERT on three dataset sizes 1000, 3000, and 5000 samples. Each of the experiments is conducted with the same conditions in terms of the maximum sequence length, batch size, and the number of epochs and is still executed solely using the CPU. This setup allows us to study the performance of models in the presence of more data systematically and allows us to estimate the scale of the smallest quantity of data that can be used to achieve a competitive level of performance.

Transformer models are based on scaled dot-product attention:

$$Attention(Q, K, V) = soft\max\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad \textbf{(4)}$$

and Q, K, V are query, key and value matrices depending on the input and dk is the size of key vectors. This mechanism allows every token to examine all the other tokens simultaneously, and then both the local and long-range dependencies are simultaneously recorded [3, 11].

3.7 Evaluation Metrics

The models were evaluated on exactly same 14,308-samples held-out test set using four standard classification metrics:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\Pr e\,cision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1 - score = \frac{2 * \Pr e\,cision * Recall}{\Pr e\,cision + Recall} \quad (8)$$

F1-score is most informative here because it equals precision and recall and is robust to the near-equal class distribution in our set.

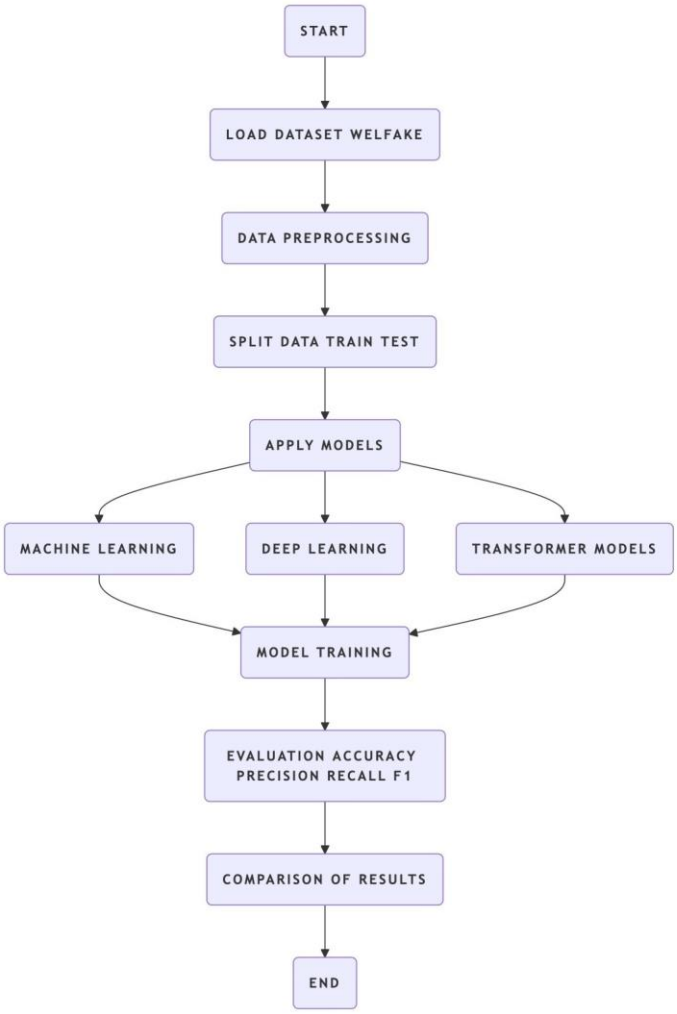

Fig. 1. Flowchart - Proposed system architecture

Figure 1 depicts the entire process used in this study, from loading the dataset to comparing results.

4. Experiments and Results

All the experiments were performed on Google Colab under Python 3 — the code relied on NumPy, Pandas, Scikit-learn, TensorFlow/Keras and Hugging Face Transformers. Hardware limits consisted of Intel Core i5 CPU with 12 GB RAM and no GPU — every model trained on the CPU only, no GPU.

The WELFake dataset is divided into two parts: 80% for training and 20% for testing. This produced 57,229 training samples and 14,308 test samples. Each split used random state=42 so that every run stayed identical and results stayed reproducible and they include multiple training sizes of DistilBERT (1000, 3000, 5000).

4.1 Overall Model Comparison

Figure 2 records the accuracy, precision, and recall besides F1-score for each of the five models. All values lie within a band of about one percent. The WELFake dataset contains few errors, holds an even split of classes and draws articles from many outlets — those properties let even simple models reach high scores once the features are well crafted.

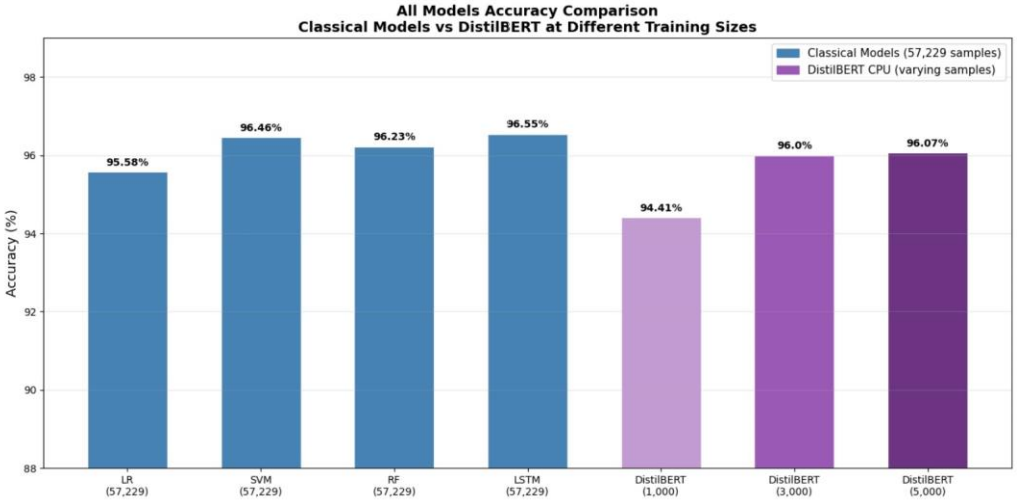

Fig. 2. Accuracy comparison of every model on the WELFake.

Table 1 lists the exact figures — LSTM tops the list with 96.55 percent accuracy and therefore ranks first overall. Among the traditional algorithms, SVM reaches 96.46 percent; Random Forest attains 96.23 percent, and Logistic Regression stops at 95.58 percent. DistilBERT yields 96.00 percent after training on 3000 records and additional experiments were conducted on 1000, 5000 samples as part of scaling, while the rest of the models saw the full set. The outcome shows that a compact transformer keeps pace with larger setups when computing capacity is restricted.

Model	Train Samples	Accuracy	Precision	Recall	F1 score
Logistic Regression	57,229	95.58%	95.10%	96.21%	95.65%
SVM	57,229	96.46%	96.00%	96.00%	96.00%
Random Forest	57,229	96.23%	96.00%	96.00%	96.00%
LSTM	57,229	96.55%	96.40%	96.79%	96.60%
DistilBERT (1,000)	1,000	94.41%	94.50%	94.42%	94.46%
DistilBERT (3,000)	3,000	96.00%	95.94%	96.31%	96.12%
DistilBERT (5,000)	5,000	96.07%	95.57%	96.69%	96.13%

Table 1. Performance of All Models on WELFake Test Set

Performance of all models on complete dataset except DistilBERT, transformer results were recorded across multiple dataset sizes to show the effect of training data scaling under CPU constraints namely 1000, 3000, 5000 samples.

4.2 Confusion Matrix Analysis

In addition to the above, the performance of the model can be understood through the analysis of confusion matrices. In the case of Logistic Regression, as depicted in Figure 3, the model was found to have performed well, as indicated by 6,723 real articles and 6,953 fake articles being classified correctly. However, there were 358 false positives, i.e., 358 real articles being classified as fake, and 277 false negatives, i.e., 277 articles being classified as real. The difference between the number of false positives and false negatives can be attributed to the fact that there is a higher possibility of real and fake articles having similar word usage, particularly in political articles.

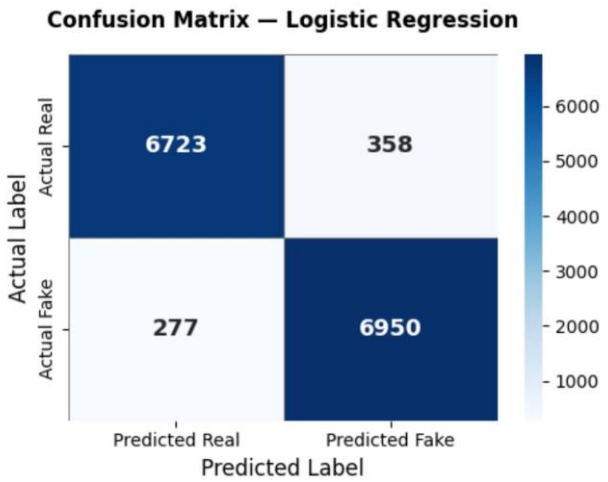

Fig. 3. Confusion matrix for Logistic Regression on the WELFake test set (14,308 samples). Labels: 0 = Real, 1 = Fake.

In the case of LSTM, as depicted in Figure 4, there were 261 false positives and 232 false negatives. The classification performance was found to be better in LSTM as compared to Logistic Regression.

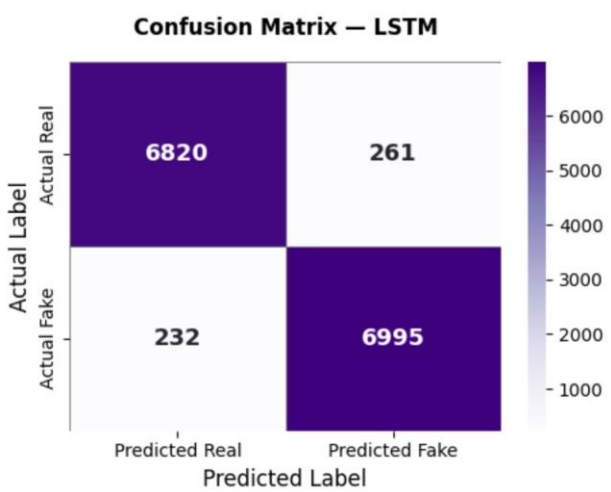

Fig. 4. Confusion matrix for LSTM on the WELFake test set (14,308 samples). Labels: 0 = Real, 1 = Fake.

4.3 LSTM Training Behavior

The LSTM model recorded the highest performance. It attained an accuracy of 96.55%, precision of 96.40%, recall of 96.79%, and an F1-score of 96.60%. Fig. 5 below shows the training and validation accuracy of the model over the course of five epochs.

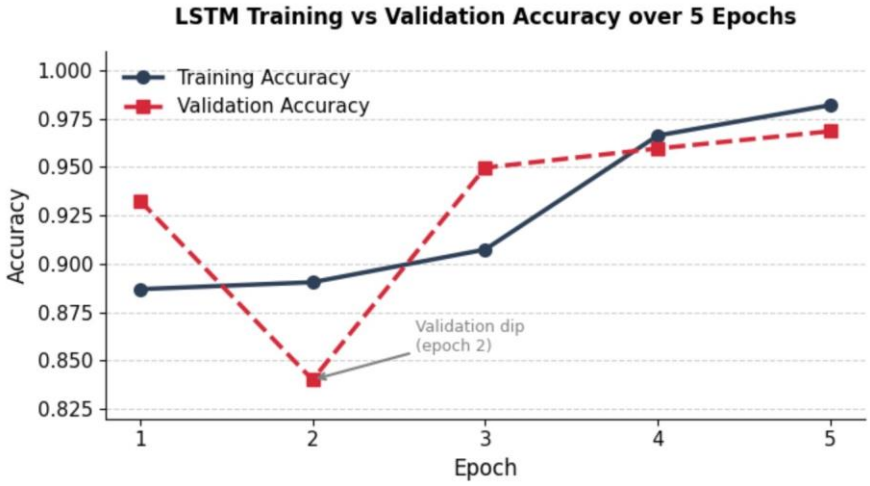

Fig. 5. LSTM training and validation accuracy over 5 epochs on the WELFake dataset.

The validation accuracy of the model experiences a decline at epoch 2. Here, the validation accuracy drops to 0.84. However, the validation accuracy increases significantly in the subsequent epochs, attaining 0.9685. On the other hand, the training accuracy of the model increases steadily. This fluctuation in the validation accuracy is a common phenomenon in recurrent neural networks. In this model, the application of a dropout rate of 0.5 also contributed to the reduction of overfitting. This is evident from the convergence of the training and validation accuracy.

4.4 DistilBERT Training Analysis

The DistilBERT model obtained a high accuracy of 96.00%, precision of 95.94%, recall of 96.31%, and a high F1-score of 96.12%. These were obtained under a constrained environment, with only 3,000 samples used for training in this configuration, as part of overall scaling analysis, a maximum sequence length of 64 tokens, a batch size of 2, and only 2 epochs on the CPU.

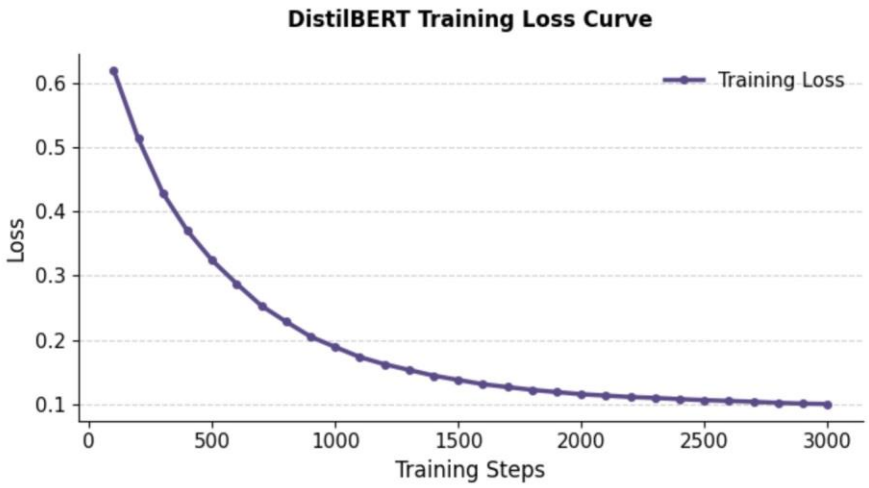

Fig. 6. DistilBERT training loss curve over 3,000 steps (2 epochs, 3,000 training samples, CPU only).

Fig. 6 shows the training loss of the model over 3,000 steps. From the plot, it is observed that the loss decreases from around 0.6 to less than 0.2. However, there are some oscillations in the loss, especially around the 800th step and the 1,300th step. This is probably because of the low batch size of the model. With only two samples per update, the model's gradients may be less stable.

The overall decreasing nature of the loss curve, however, shows that the model is learning well. It is also observed that the DistilBERT model's performance is comparable to the Logistic Regression model's performance, despite being trained on smaller subset of data, which used the full dataset. This shows the effectiveness of the pre-training of the transformer model. In the DistilBERT model, much of the language understanding is already built in.

4.5 DistilBERT Data Scaling Analysis

The DistilBERT transformer model was evaluated on three different sizes of dataset namely 1000, 3000, 5000 under identical training conditions. On 1,000 samples it achieved accuracy of 94.41% which increased to 96.00% on 3000 and then to 96.07% on 5000 samples. Notably, 5.2% of complete dataset samples are 3000 samples, achieved close performance to SVM which was trained on all 57,229 samples, with only a minute gap. This shows data efficiency of DistilBERT under CPU constraints.

Fig 7 shows steep improvement from 1,000 to 3,000 samples and then makes a steady plateau between 3,000 and 5,000 samples. This shows that pretrained knowledge in DistilBERT compensates need for large data and after a certain specific size it learns automatically commonly used patterns in the datasets. So, 3,000 samples can be considered a threshold value or critical value where pre-training is effectively utilized.

This hold a strong practical importance as students and researchers do not have access to high end GPUs and have limited labeled data. But the results showed that DistilBERT with 3,000 samples performed well and with increase in sample size the performance improves slightly. At 5,000 samples DistilBERT outperformed logistic regression and random forest which were trained on full dataset (57,229 samples). In a real world deployment scenarios with resource constraints, a modestly fine tuned data along with a pre-trained language representations will be a viable and practical solution.

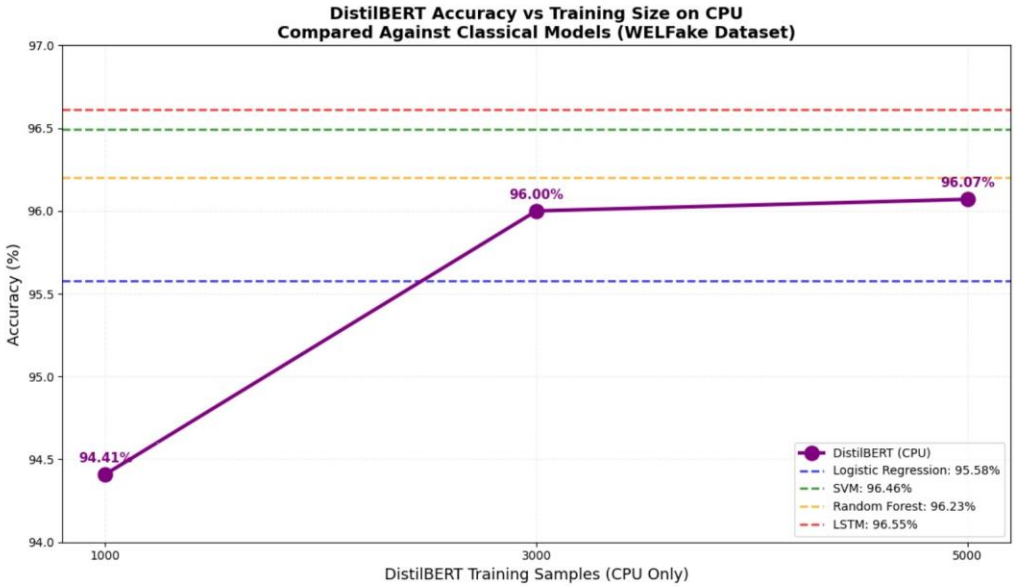

Fig. 7. DistilBERT accuracy vs training size under CPU constraints, compared with classical models trained on the complete dataset.

5. Discussion

From the experiments performed, it is clear that the performance gap between classical models, namely SVM 96.46% accuracy and neural models LSTM 96.55% accuracy, is relatively small. The WELFake dataset's articles carry various word patterns and sequences which even base level models perform with ease at the same power with complex models which further closes this gap [6].

A major observation is that the transformer DistilBERT performed almost at the same level as that of other models, despite limited data and resources. It performed on reduced sequence length and smaller batch size. Despite these restrictions, the model performed at the same level as that of other models trained on a full dataset. It can be inferred that transformers need not depend on large scale computation to generate results, it can still perform if the pretrained representations are harnessed properly.

With accordance to our knowledge this is the first systematic scaling analysis of DistilBERT on WELFake under CPU only constraints. Scaling analysis of DistilBERT was done on 1,000, 3,000, 5,000 samples and with increase in data samples, the performance of the model also improved. Notably, the model trained on 3000 samples roughly 5.2% of the total dataset; a marginal gap was there between transformer and SVM. This shows that transformer models are highly data efficient and can achieve considerable performance on limited dataset and they may offer a better return on annotation effort than classical models [4, 5].

By not depending on default preprocessing configurations TF-IDF settings, we set ngram_range=(1,2) with max_features=10000 and max_df=0.7 which improved the Logistic Regression performance. This shows that by changing preprocessor configurations in every scenario gives a better variable result than just using the default ones.

6. Conclusion

This paper compared three generations of NLP models which were used to detect fake news namely — classical machine learning, LSTM deep learning and DistilBERT transformer on WELFake dataset [12]. The results showed that it is not necessary for transformers to win every time.

LSTM scored the overall highest accuracy at 96.55%, followed by SVM at 96.46%, Random Forest at 96.23%, Logistic Regression at 95.58% and DistilBERT at 96.00% when trained on 3,000 samples. Our scaling analysis showed that at 5,000 samples DistilBERT got 96.07% which exceeded Logistic Regression and Random forest on complete dataset. One could see that the margin between the best and worst performing models is less than 1% because we used a high-quality and balanced dataset. If we were to use a noisier and more field specific data the gaps would have been bigger.

DistilBERT was trained across three sample variations and it was observed that it learns better when data is increased. On 1,000 samples, accuracy was 94.41%, on 3,000 samples it was 96.00%, and on 5,000 samples it reached 96.07%. A marginal difference of 0.46% was observed between SVM trained on 57,229 samples (complete dataset) and DistilBERT. Thus, it shows that transformer models use robust pre-trained representations which improve their learning efficiency even with limited data. This proves that high-end models can be used in low-resource environments like academic research, where high-end GPUs are not available. These results give a practical idea as to which model configurations one can choose when resources are limited.

The transformer results are highly significant. Even with CPU constraints utilizing 1,000, 3,000 and 5,000 samples, with sequences truncated to 64 tokens and a batch size of 2, the model performed in close proximity to full-data trained models. This shows the power of pretrained models in low resource settings and computational constraints can still produce meaningful results, when disclosed honestly, can rather be an informative finding which people usually choose to hide [5].

Classical models like SVM and Random Forest are fast to train, require no GPU and perform close to advanced newer models. They are a good choice in a low-resource environment and provide decent interpretable results. LSTM sits at middle ground between classical models and transformers as they are easier to train than transformers and still give competitive accuracy [7].

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